iLearn Practice Exam

10BR · Chapter / Topic 5

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Chapter 5: The Internet and its Uses

Name: _____

Class: _____

Date: _____

Time allowed: 55 minutes

TOTAL MARKS: 46

Marks obtained: ..... / 46

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 46.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 5 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.
· Only 46 marks of past-paper questions are available for this chapter, so the recommended time is 55 minutes (1 min 1

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 A company has a website that is suffering a distributed denial of service (DDoS) attack.

(a) Draw and annotate a diagram to show the process of the DDoS.

[5]

(b) Identify a solution that can be used to help prevent the DDoS attack being successful.

..... [1]

2 A company has a website. Users use the internet and the world wide web to access the website.

(a) Describe the difference between the internet and the world wide web.

.....

.....

.....

..... [2]

(b) The website has a uniform resource locator (URL). The URL has three different parts.

Identify the **three** different parts that are included in the URL.

1 .....

2 .....

3 .....
[3]

(c) One function of a web browser is to provide an address bar for a user to enter a URL.

Give **three** other functions of a web browser.

1 .....

2 .....

3 .....
[3]

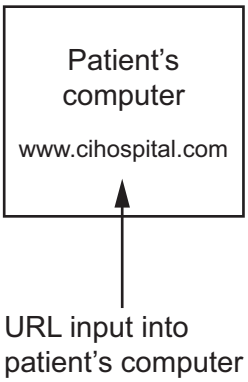
BLANK PAGE

3 A hospital has computers and technology that it uses when caring for its patients.

(a) Patients use a web page to log into their account to pay for the treatment they have received in the hospital.

(i) The patient enters the uniform resource locator (URL) for the web page into their browser and the web page is requested and displayed.

Complete and annotate the diagram to show how the web page is requested for the patient.



[6]

(ii) A patient is making a payment through a secure connection.

The secure connection is created using the secure socket layer (SSL) protocol.

Explain how a secure connection is created using the SSL protocol.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

4 Web pages need to be located and retrieved to be displayed on a device.

(a) Complete the paragraph about locating and retrieving web pages.

Use the terms from the list. Some of the terms in the list will **not** be used.

You should only use a term once.

certificate	database
domain name server (DNS)	encryption
hypertext markup language (HTML)	internet protocol (IP) address
low-level language	protocol
request	uniform resource locator (URL)
web browser	web server

The user types the URL into the address bar. The web browser sends the domain name from the URL to the ..... .

This searches for the matching ..... and sends this back to the ..... .

The web browser sends a ..... to the web server for the web page. The web server then sends the data for the web page back to the web browser. The web browser renders the ..... to display the web page.

[5]

(b) The transmission of data between the web browser on a device and a web server can be secured using the secure socket layer (SSL) protocol.

Complete and annotate the diagram to show the process of sending the data using SSL.



[5]

5 Cyber security threats are a problem when using the internet.

(a) Complete the table by writing the missing terms and descriptions about cyber security threats.

Each term and description must be different.

term	description
....	An attack where a third party repeatedly tries to crack a password by entering different combinations.
hacking	....
....	An attack where multiple computers send a large number of requests to a web server with the aim of crashing it.
spyware	....
....	A security threat where software is used to analyse data packets during transmission. Any packets that look useful are sent to a malicious third party.

[5]

(b) Explain how access levels are used to help prevent a cyber security attack being successful.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

END OF QUESTION PAPER